The VFR times were based on schedule block times for the New York Airways' VFR operation, with ground taxi time substracted to yield actual flight times. The IFR times were based on the average point-to-point results obtained during these tests plus an allowance of 40 seconds to compensate for the fact that the tests were initiated at an altitude of 30 m (100 ft) and terminated at an altitude of 61 m (200 ft). As might be expected, the flight time for the IFR task was substantially greater than for the VFR task, which is characterized by direct terminal-to-terminal flight with minimum constraints. By contrast, the IFR procedures resulted in increased stage lengths because it was necessary to operate in the specified route corridors and to cope with airspeed and altitude constraints. The airspeed constraint required deceleration to the low approach speed while still several miles from the landing pad, and the altitude constraint required additional time for climbing to and descending from higher minimum en route altitudes. The total time for the IFR task was about 55 percent greater than for the VFR operation, the greatest loss occurring between Morristown and Newark because of the required procedure turn which took approximately 5 min. A second MLS at Newark would have eliminated this delay. These time differences, however, do not make any allowance for ATC-related delays, which would be most severe, of course, during IFR conditions.

Pilot Workload

An actual IFR operation of the type investigated in this study would typically require a pilot-copilot team to handle the normal tasks including navigation, aircraft control, and ATC communications. Because of simulator constraints, however, the operation was conducted by a single pilot, but the workload was reduced accordingly by eliminating the ATC-communications task and by requiring the radios to be tuned prior to each run. Discussion of pilot workload in this paper, therefore, relates primarily to the navigation and control tasks.

The workload in flying the circuit varied from very low to extremely high. The low workload condition occurred during the en route portion of the task while tracking a station radial at constant speed and constant altitude. The highest workload occurred during the final approach and could range from moderate to extremely high, depending on how well the pilot had been able to establish the desired conditions of speed, crab angle, and power prior to glide-slope intercept.

Sensitivity of both the localizer and the glide slope caused difficulty at short range. With the localizer and glide-slope transmitters colocated at, or near, the landing pad, small linear deviations at short ranges caused large instrument deflections, which gave the pilot the false impression that tracking performance was very poor. This problem could be eliminated, of course, by beam softening to prevent increased sensitivity with reduced range.